RevIN Experiment 1: Normalization Components

1 Theoretical overview and goal

Global standardization uses training-wide statistics; instance normalization uses the current look-back. The implementation selects its location and scale independently:

$$\ell_x \in \{0, \mathrm{mean}(x), x_L, \mathrm{median}(x)\}, \qquad s_x \in \{1, \mathrm{std}(x), \mathrm{MAD}(x)\}.$$

Removing either statistic can reduce distribution shift but can also discard predictive information. MIN additionally unties RevIN's input and output affine transformations:

$$\tilde{x} = \gamma \frac{x - \ell_x}{s_x} + \nu, \qquad \hat{y} = s_x \left[\alpha \frac{f_\theta(\tilde{x}) - \nu}{\gamma} + \beta \right] + \ell_x.$$

These four global per-channel parameters are identity-initialized. MIN is non-personalized: it has no cluster parameters or empirical initialization.

Experimental goal. Isolate location, scale, robust statistics, symmetric affine parameters, and global output modulation under matched losses.

2 Implementation details

Family	Location/scale	Extra parameters
`none`	none	none
`standard`	training-global mean/std	none
`mean_only`	window mean/none	none
`scale_only`	none/window std	none
`instance`	window mean/std	none
`median_mad`	window median/MAD	none
`revin`	window mean/std	symmetric affine
`min`	window mean/std	untied input/output affine

Each family is trained with MSE and nMSE; concrete method names append `_mse` or `_nmse`. Statistics are computed from training data (`standard`) or the current window, inverted after forecasting, and evaluated using original-space MSE.

Small uses five datasets and the three shared default settings; full uses nine datasets and five settings. Publication runs use three seeds, 10,000 optimizer steps, validation every 1,000 steps, and horizon-strided evaluation. Full uses PatchTST; ultra adds DLinear. Test is an isolated one-seed, 2,000-step gate.

Code: `src/utils/models.py`; **launcher:** `revin.slurm`. The launcher is a resumable `train,tables` workflow whose separate stage implementations live under `src/slurm/`.

Run `test`, `small`, `full`, then `ultra`; publication profiles reuse completed seeds. Grouped tables report mean ± seed standard deviation with explicit row multipliers.

3 Interpretation

Mean/scale-only isolate statistics; instance–median/MAD tests robustness; instance–RevIN tests symmetry; RevIN–MIN tests untied global modulation.